

Week 4

Bernoulli, continuity, and their limits

Automotive Aerodynamics Independent Study

Industrial Design × Physics (DePaul)

Physics mentor · 10-week IS

Today's agenda

1. Recap: q is kinetic energy per volume; $F = qCA$
2. Continuity: why a narrower path speeds the stream up
3. Bernoulli: $P + \frac{1}{2}\rho v^2$ along one streamline
4. Worked ΔP , including a diffuser
5. Where the idealization fails: separation and wakes
6. The path-length slogan
7. Honest captions for the wing and the diffuser
8. Deliverables

Learning targets (by end of Week 4)

You will be able to:

- State incompressible continuity: $A_1 v_1 = A_2 v_2$ along a streamline tube
- Write Bernoulli's equation and compute a sample ΔP at constant height
- Say which assumptions fail for a stalled spoiler or a separated diffuser
- Write an honest caption: marketing sentence and physics sentence, side by side

Recap — what still does the forcing

Weeks 1–3 still govern the car:

$$F \sim \Delta P A, \quad \Delta P = C q, \quad q = \frac{1}{2} \rho v^2$$

q is the kinetic energy in each cubic meter of oncoming air.

C is how much of that scale this shape turns into a net force.

Week 4 asks a narrower question: **along one tidy streamline, how are P and v related, and when is that story the wrong story?**

Learning outcome LO4 (syllabus)

Critique Bernoulli-only explanations; identify separation and wake effects qualitatively.

YouTube will say “fast air is low pressure.”

That sentence is a **special case**, not a law of cars.

Use it where the assumptions hold. Measure C_D and C_L where they do not.

Two conservation statements

Statement	What is conserved	Needs
Continuity	mass per time through a tube	density about constant
Bernoulli	a mechanical-energy budget	steady, inviscid, along a streamline

Continuity tells you the **speed**.

Bernoulli turns that speed into a **pressure change**, if the flow stays ideal.

Continuity (incompressible)

Air has density ρ . Through a cross-section of area A at speed v , the mass flow is

$$\dot{m} = \rho A v$$

If ρ is the same at two stations on one stream tube, and no air leaks through the walls,

$$A_1 v_1 = A_2 v_2$$

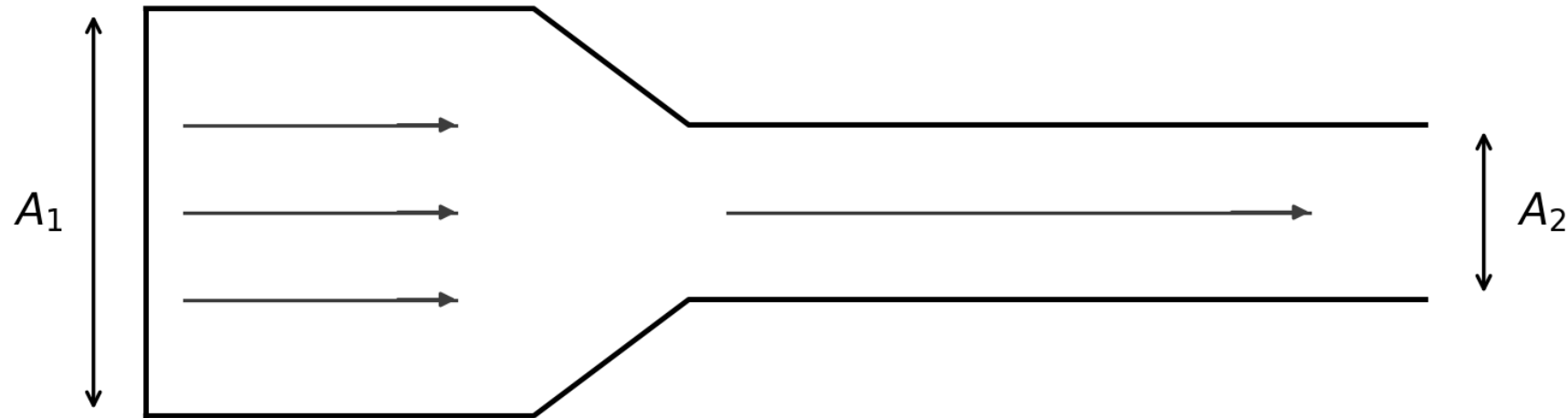
Narrower $\Rightarrow$ faster. Wider $\Rightarrow$ slower.

Continuity as a picture

Continuity: a narrower tube speeds the same stream up

slower v_1

faster v_2



incompressible: $A_1 v_1 = A_2 v_2$

Worked continuity

$$A_1 = 0.020 \text{ m}^2, v_1 = 10 \text{ m/s}, A_2 = 0.010 \text{ m}^2, \rho = 1.2 \text{ kg/m}^3.$$

$$v_2 = \frac{A_1 v_1}{A_2} = \frac{(0.020)(10)}{0.010} = 20 \text{ m/s}$$

Mass check, both stations:

$$\dot{m} = (1.2)(0.020)(10) = (1.2)(0.010)(20) = 0.24 \text{ kg/s}$$

Halving the area doubled the speed. The kilograms per second did not change.

Bernoulli, with the assumptions in view

For **steady, inviscid, incompressible** flow **along a streamline**:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Read the middle term as q on that streamline, not as the free-stream q of Week 2 unless this streamline *is* the free stream.

Assumptions, said out loud:

1. Steady (or a fair time average)
2. Inviscid: viscosity is not eating mechanical energy
3. Incompressible: ρ constant, which continuity already used
4. Along **one** streamline, not from a random point to another
5. No fan or pump doing work on that streamline between the two stations

Drop the height term for a car

Between two points a few centimeters apart in height,

$$\rho g \Delta h \approx (1.2)(9.8)(0.05) \approx 0.6 \text{ Pa}$$

Speed changes in these problems move P by **tens or hundreds of pascals**.

For a quick car estimate, same height:

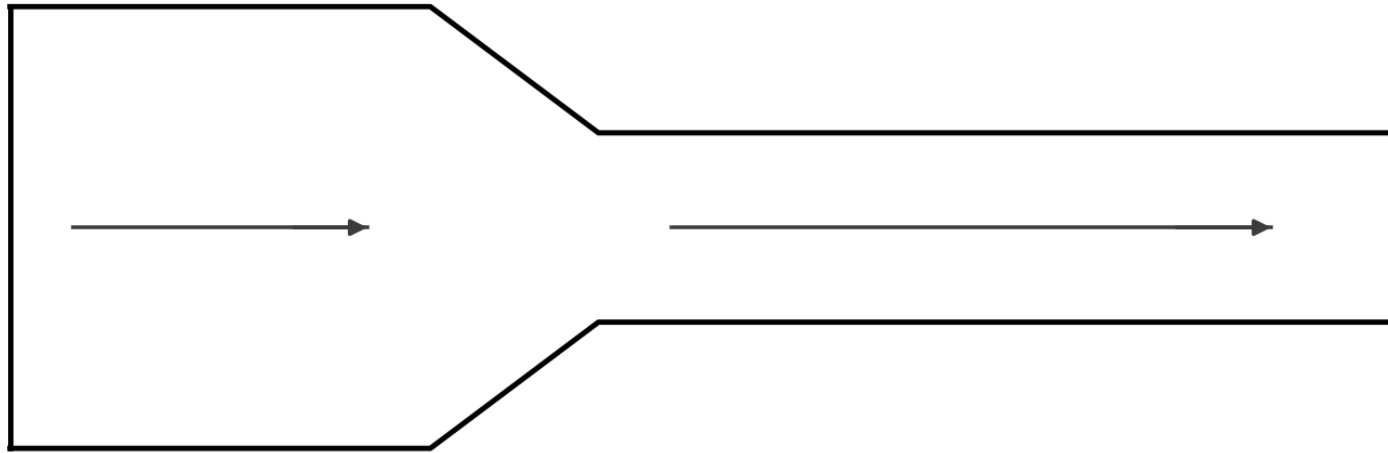
$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

Bernoulli stations

Bernoulli (ideal): speed up along the streamline, pressure falls

slower, higher P

faster, lower P



same height, along one streamline: $P + \frac{1}{2}\rho v^2 \approx \text{const}$

Worked ΔP (checkpoint 1)

Speed along the streamline goes from 10 to 20 m/s. $\rho = 1.2 \text{ kg/m}^3$.

$$P_2 - P_1 = \frac{1}{2} \rho (v_1^2 - v_2^2) = 0.6 (100 - 400) = -180 \text{ Pa}$$

Pressure is **lower in the faster region**, by 180 Pa, inside this idealization.

Compare with Week 2: free-stream q at 20 m/s is 240 Pa. A 180 Pa change is a large fraction of that scale, which is why C can be order 1.

Checkpoint 1 — how to say it

Air in a duct speeds from 10 to 20 m/s. Estimate ΔP . Where is pressure higher?

$$\Delta P = P_{\text{fast}} - P_{\text{slow}} = -180 \text{ Pa}$$

Pressure is higher where the air is **slower**.

Units: pascals. Direction: from the high- P face toward the low- P face if this ΔP acts across a surface (Week 2).

A diffuser, Bernoulli-only

A diffuser is a channel that **opens** in the flow direction. Continuity:

$$A \text{ up} \Rightarrow v \text{ down}$$

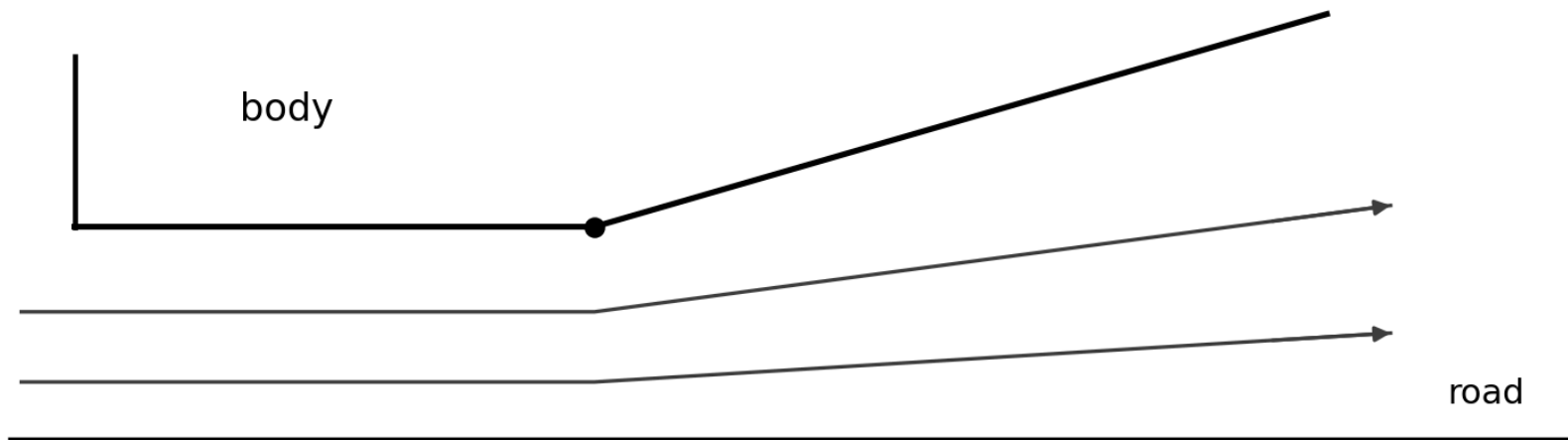
Bernoulli, if the flow stays attached and ideal:

$$v \text{ down} \Rightarrow P \text{ up}$$

That rise is **pressure recovery**. Under the car, a higher pressure toward the rear of the underbody is part of how a diffuser is *supposed* to help downforce: the underbody can run at lower P upstream of the recovery.

Diffuser cartoon

Diffuser cartoon: slow the underbody flow and recover pressure
if attached: area up, v down, P up



A steep kink can separate. Pressure recovery is a hypothesis until the flow stays attached.

Worked recovery

Attached diffuser, area doubles, speed falls from 20 to 10 m/s:

$$\Delta P = 0.6 (400 - 100) = +180 \text{ Pa}$$

That +180 Pa is the ideal gift.

A ramp that is too steep separates. Then the streamline you used **leaves the wall**, the ideal ΔP is not delivered, and the wake is the real object.

Week 8 measures ride height and diffuser shape. Today: the sign of the ideal story, and the condition "if attached."

Checkpoint 2

A diffuser slows underbody flow as area increases. In a Bernoulli-only story, what happens to pressure? What is “pressure recovery”?

Pressure **rises** as the stream slows.

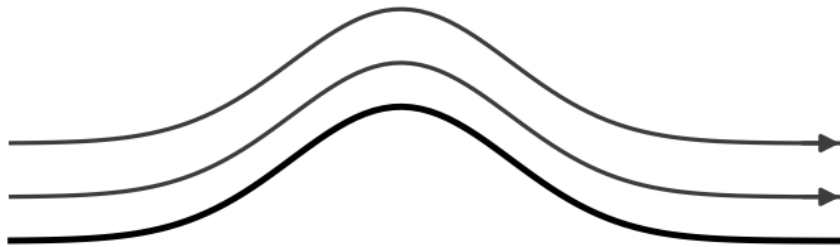
Pressure recovery is that rise, back toward the pressure outside the channel.

Add one sentence the slogan forgets: recovery happens **if the flow stays attached**. A separated diffuser is a wake, and this ΔP is then a hypothesis you did not earn.

Attached flow and a wake

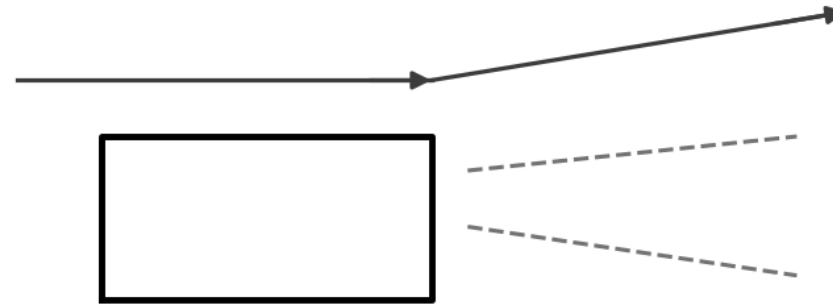
Attached flow can follow a gentle surface; a sharp corner sheds a wake

attached



streamlines follow
the surface

separated



wake

streamlines leave;
Bernoulli on the wall fails

How to read that figure

Left: the surface is gentle, streamlines follow it, and a Bernoulli estimate along one of them can be a fair sketch.

Right: at a sharp corner the near flow leaves. Downstream is a **wake**: slow, messy, often unsteady. The wall behind the corner is not a streamline anymore.

Why “fast air = low pressure” fails in a wake

Bernoulli compares two points **on the same streamline**.

In a wake:

- The air next to the surface did not get there by a tidy acceleration along that surface
- Viscosity and separation have already spent mechanical energy
- A low speed in the wake does **not** mean “therefore high pressure by Bernoulli”

Bluff bodies, sharp spoilers, and stalled wings make force mostly with a **pressure difference between the front and the wake**, which is the Week 3 coefficient story. It is not a single Bernoulli evaluation on the back face.

The path-length slogan

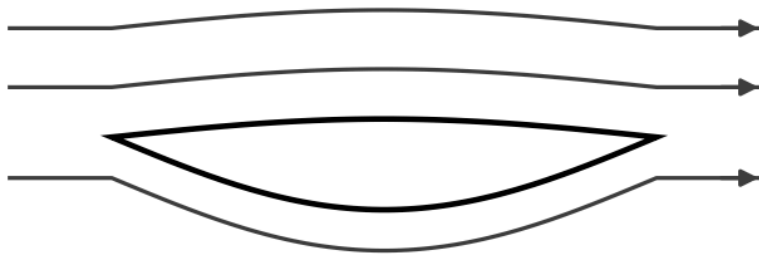
You will hear: “air over the long path must go faster so the two streams meet at the trailing edge, so the pressure is lower.”

Equal transit time is **not** a law. The upper and lower streams do not have an appointment at the trailing edge.

Slogan and the safer picture

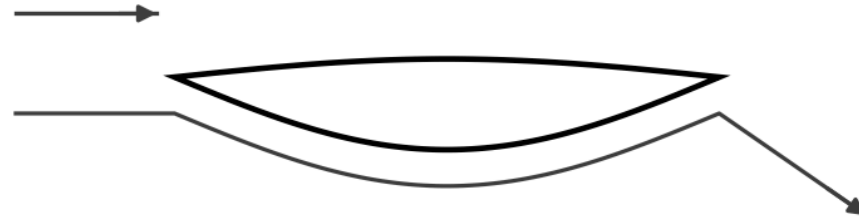
Downforce is a pressure force from turning the air, not a path-length race

common slogan



“longer path, so faster”
Equal transit time is not required.

safer picture



the wing turns the flow;
net force is still ΔP

Safer picture, already in Weeks 1–3: the wing **turns** the air. The net force is ΔP on the surfaces. C_L is how much of qA that turn produces.

Checkpoint 3

Give one reason a real rear wing's downforce is not fully explained by "air travels farther over the top."

Any one of these is enough:

- The two streams need not take equal time
- If the wing stalls, the "long path" is not a surface streamline
- The force is the pressure distribution from turning the flow, which you will **measure** as

$$F_L = qC_L A$$

A longer camber line can be a design choice. It is not, by itself, the derivation of downforce.

Checkpoint 4 — *Stretch*

List three assumptions of Bernoulli and mark which fail for a sharp-edged spoiler in a stalled condition.

Assumption	Stalled sharp spoiler
Inviscid	Fails. Separation is a viscous event
Along the surface streamline	Fails. The surface streamline has left
Steady	Weak. The wake fluctuates; a time average is the best you get
Incompressible	Usually still acceptable at these speeds
ρgh negligible	Still fine; height is not the problem

Honest captions

A caption has two sentences.

Wing

- Marketing: "The inverted wing presses the car into the road."
- Physics: "If the flow stays attached, the wing turns air toward the road and the surface ΔP is a downward force. If it stalls, that map changes and C_L must be measured, not assumed from a path length."

Diffuser

- Marketing: "The diffuser seals the floor and adds grip."
- Physics: "If the underbody stream slows while attached, Bernoulli gives pressure recovery. Ride height, the ramp angle, and separation decide whether that recovery happens. Week 8 tests it."

Where to look for separation on *your* body

Mark likely edges on the side view, as hypotheses:

- Hood to windshield
- Roof to rear glass
- Sharp spoiler or wing trailing edge
- Start of a diffuser ramp (the kink in the figure)
- Exposed wheel arches and mirror stalks, if the model has them

Each mark needs the sentence: "I will believe this if the tufts (or the load cell) show ..."

Tufts and smoke are optional later. The sketch is due this week.

Course stance

Use continuity and Bernoulli to estimate **attached** speeding-up and slowing-down.

Then ask, in this order:

1. Is the flow attached, or is there a wake?
2. Am I still on one streamline?
3. Did I ignore viscosity that is actually doing the separation?

If the answer is a wake, stop the Bernoulli arithmetic and go back to $F_D = qC_D A$ and $F_L = qC_L A$.

Design / build tasks this week

- ☐ Honest caption for the wing
- ☐ Honest caption for the diffuser (or the underbody, if the diffuser is not drawn yet)
- ☐ Separation-edge sketch on the body
- ☐ Checkpoint 1–4 with units

Physics before a more dramatic CAD surface: a deeper ramp with no attached-flow sentence is not yet a Week 4 artifact.

Deliverables checklist

Deliverable	Where
Checkpoint 1–4	notes or <code>curriculum/week-04-checkpoint.md</code>
Two honest captions	<code>cad/notes/</code> or <code>docs/</code>
Separation-edge sketch	with the captions

Due before the Week 5 midterm review.

How you will be assessed on Week 4

Strong: ΔP with units and a sign; assumptions named; captions that say when the story stops.

Weak: “fast air = low pressure” on a stalled spoiler; a diffuser claim with no “if attached”; path length offered as a derivation.

Looking ahead — Week 5

Reynolds number. The same shape at a much smaller vL can separate in a different place, so a model C_D is not a full-car C_D .

Bring a frozen model scale and a chosen length L . Week 5 is also the midterm design review.

Instructor talking points

- Praise the curiosity in a Bernoulli slogan, then ask “which streamline?”
- If the student evaluates Bernoulli inside a wake, redirect to the separated-flow figure.
- Keep ground-effect and diffuser tuning for Week 8; this week is the ideal sign and the failure mode.
- The height term is a good check that not every term in an equation matters at car scale.

References for this lecture

- Course note: [physics/04-bernoulli-and-limits.md](#)
- Week plan: [curriculum/week-04-bernoulli-and-limits.md](#)
- q and ΔP : [physics/02-pressure-and-dynamic-pressure.md](#)
- Coefficients: [physics/03-drag-and-lift.md](#)
- Figures: [curriculum/slides/figures/](#)

End of Week 4 lecture

Next actions

1. Finish the checkpoint with units
2. Write two honest captions
3. Sketch separation edges
4. Freeze a model scale for the Week 5 review

Questions?